

TERRA Product Features by Tier

João Leonardi^a

^a*Computer Vision Research Center, iCEV Institute of Higher Education, Teresina, Brazil.*

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Abstract

This document lists eleven candidate products for TERRA in three tiers. Tier 1 holds the three items that complete the current workflow (area, fields, jobs, results) or were requested directly. Tier 2 holds the five items that the product comparison brought to this project lists and TERRA lacks. Tier 3 holds three extensions proposed during that analysis. Each item states what it delivers, the code it builds on, what it depends on, and a qualitative complexity (low, medium or high). The complexity is reported for every item and does not determine the order. Within a tier, items are ordered by dependency.

1 Current state

Implemented. Classification with per-pixel confidence; comparison with MapBiomass; NDVI series with phenology (start, peak and end of season, computed over a run's area); surface water; mineral map; field delineation; the job queue (`internal/jobs`); and the single compositor graph for every field of an area (`frontend/src/lib/fieldSets.ts`).

Not implemented. Sentinel-1, Landsat or HLS imagery; error-adjusted area estimation; training on user samples; management zones; socio-environmental checks; vegetation anomaly. Surface water is the only product that computes an anomaly.

2 Tier criterion

Tiers are assigned by where a request comes from and what it unblocks, not by effort:

Tier 1 Completes the workflow already built (area → fields → jobs → results), or was requested directly. Items in later tiers report through these.

Tier 2 Listed in the product comparison brought to this project and absent from TERRA.

Tier 3 Proposed during the analysis and listed in neither source.

Within a tier, an item precedes the items that depend on it. Where no dependency separates two items, the order follows the reach of the result: products usable across all fields before products for one use. The complexity column is a judgement from reading the code; it states scope and gives no schedule.

3 Tier 1: completing the current workflow

The per-field table comes first because every later per-field product needs a place to report one row per field. The season dates are already computed in every field job and need only that table. Vegetation health was requested directly and is a new job kind that the queue and the single graph accept.

#	Product	Delivers	Builds on	Depends on	Complexity
1	Per-field table	One row per field (area by class, season dates, mean confidence), exportable to CSV.	Per-field runs; the single compositor graph (<code>fieldSets.ts</code>).	None.	Low
2	Season dates per field	Emergence, peak and senescence for each field, with an uncertainty window.	<code>phenology.py</code> : Savitzky–Golay smoothing and a 50 % amplitude threshold, the rule TIMESAT applies (Jönsson and Eklundh, 2004). It already runs in every field job.	Item 1 (display across fields).	Low to medium
3	Vegetation health	NDVI/NDRE anomaly of each field against its own history and against neighbouring fields of the same crop, as a map and a score per field.	Sentinel-2 series and red-edge bands B05–B07; a new job kind (<code>jobWork</code> , <code>JOB_KINDS</code>).	Item 1 (score per field); decision D1.	Medium

4 Tier 2: gaps against the product comparison

Socio-environmental checks lead the tier because they depend on no model and apply to every field. The error-adjusted area follows because its labelling screen is the one the training on user samples reuses. Management zones use only the field polygons and the series. Sentinel-1 closes the tier: it feeds the season dates and vegetation health of Tier 1 under cloud and is not a prerequisite for any other item in this tier.

#	Product	Delivers	Builds on	Depends on	Complexity
1	Socio-environmental checks	Overlap of each field, in hectares and with dates, with PRODES/DETER (deforestation after 2008, Brazilian Forest Code; after 31 Dec 2020, EUDR), CAR/SICAR (APP and legal reserve), IBAMA embargoes, indigenous lands and conservation units.	Field polygons; area geometry. Missing: per-area download of the public layers (TerraBrasilis WFS) and the vector intersections.	None.	Medium
2	Error-adjusted area	Area of each class with a 95 % confidence interval, and user’s and producer’s accuracy (Olofsson et al., 2014).	Classification runs. Missing: stratified random sample, a labelling screen over the imagery, the estimator. The method is cited in <code>mapbiomas.py</code> and not computed.	None.	Medium to high
3	Management zones	Three to five zones per field from several seasons of NDVI/EVI (fuzzy k-means; Fridgen et al., 2004), exportable to GeoJSON or shapefile for variable-rate application.	Field polygons; Sentinel-2 series.	Multi-season series per field.	Medium

#	Product	Delivers	Builds on	Depends on	Complexity
4	Training on user samples	A classifier fitted to the region, with labelled fields (“soybean”, “maize”) as training samples.	Field polygons; the Random Forest path. Prithvi is used frozen. Missing: labelling screen, retraining.	Item 2 (labelling screen).	Medium (Random Forest) to high (network)
5	Sentinel-1 (radar)	A VV/VH series independent of cloud: season cycle in the summer crop, harvest detection from the drop in VH, water mapping under cloud.	Planetary Computer RTC collection, served through the same STAC as Sentinel-2. Missing: a sidecar slice.	None.	High

5 Tier 3: extensions proposed in the analysis

Event alerts read the series that vegetation health builds, so they follow it. HLS densifies the series that the dates and the health products read. Yield closes the list because it depends on data the user supplies.

#	Product	Delivers	Builds on	Depends on	Complexity
1	Field event alerts	Abrupt drops in the series (hail, frost, lodging, early harvest) found by break detection (BFAST; Verbesselt et al., 2010).	Per-field series; can be part of vegetation health.	Tier 1 item 3.	Medium
2	HLS (Landsat + Sentinel-2)	A revisit of two to three days, which narrows the uncertainty of season dates and health in cloudy regions.	NASA LP DAAC; the Earthdata token already used by the mineral map.	Tier 1 items 2 and 3 (consumers).	Medium
3	Yield	Calibrated only with the user’s harvest data (yield monitor); without it, a relative potential map.	Per-field series and season dates.	User harvest data.	High

6 Open decisions

D1 Reference for the vegetation anomaly: the field’s own history, neighbouring fields of the same crop, or both. The choice sets the data each health job reads.

7 Limitations

- **Criterion.** Tiers reflect the state of the code, the direct requests and the product comparison available when this list was written. A new request or a change in scope reorders them.
- **Complexity.** Qualitative, from reading the code; no effort was measured, and no item was prototyped.
- **Field accuracy.** Every per-field product inherits the field delineation, whose accuracy over western Paraná is unmeasured: no hand-digitised reference fields exist for the region.
- **Season dates.** NDVI marks emergence and senescence, not sowing and harvest. Sowing precedes emergence by several days and harvest follows senescence; with a five-day revisit and cloud, the

uncertainty is days to weeks.

- **Public layers.** PRODES, DETER, CAR and the embargo lists change on their own schedules. A check is valid for the date its layers were read, and has to state that date.
- **Yield.** Without the user’s harvest data, no absolute yield figure is supported.

References

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